

Urgency Builds Trust: A Voice Agent's Emotional Expression in an Emergency

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ABSTRACT

Voice agents play a crucial role in providing verbal guidance to users, especially during emergency situations to ensure their successful evacuation. While the vocal presentation of emotion significantly impacts communication effectiveness in human-agent interaction, the context-dependent impacts of emotional voice in an emergency setting remain unexplored. This study investigates the effects of voice agents' emotional expressions conveyed through tone (i.e., Calm vs. Urgent) during a fire emergency simulation. Participants ($N=30$) conducted an online simulation game where a voice agent instruct them on how to evacuate from a building, while their evacuation time was tracked. After the game, participants evaluated the task experience and the voice agent's trustworthiness and context-appropriateness. Results show that the voice agent using an urgent voice increased the participants' cognitive trust in the agent and context-appropriateness to the emergency, compared to the agent using a calm voice. The study findings provide design implications for a voice agent's emergency instructions, aiming for an effective evacuation process.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

Voice Agent; Emotional Expression; Trust; Emergency Simulation; Human-Agent Collaboration

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1 BACKGROUND

Virtual agents have been widely used in many fields and have demonstrated their usefulness in providing guidance. With advanced natural language processing (NLP) technologies, a speech-based agent can engage in natural conversation with users, enabling hands-free interactions. Recent studies actively adopted the voice agent in the emergency management system where the agent provides evacuation guidelines in a timely manner (e.g., [3, 10]). In an emergency situation, an agent instructs the users to take appropriate actions and escape the emergency successfully. For instance, Gan et al. [8] proposed an indoor evacuation guidance system with a virtual agent that can effectively guide users to evacuate quickly and safely in a complex museum building. Another study developed a real-time fire emergency system that navigates and tracks the users' location and provides a voice-based command to support better decision-making [21].

When delivering voice instructions, the agent should use appropriate speech tones for effective communication. A study on the Japanese tsunami warning system emphasized that the inefficient tone of warning voice is one of the reasons for the high death rate in the earthquake [2]. In particular, when emergency instructions are delivered through voice, the voice tone can influence the users' perception of the instructions and the agent. A strong body of research on Human-Robot Interaction indicates that the agent's vocal expression affects users' judgment of the trustworthiness and appropriateness of the agent [7, 18]. Therefore, examining which tone of voice should be incorporated into the emergency agent system to be perceived as effective and appropriate is critical.

Voice displays the speaker's emotional states and intentions through paralinguistic cues, such as pitch, volume, tone, speech rate, and intonation [9]. Since the perceived emotion is context-dependent, the agent's tone of voice that sounds appropriate in the emergency varies depending on the user's situation. Earlier studies highlight that emergency instructions should convey a sense of urgency in order to persuade people to evacuate immediately [2]. To convey urgency, different linguistic strategies have been employed, such as adding signal words that display assertiveness [22] or danger [14], and adjusting paralinguistic parameters including pitch and speed [5]. Regarding paralinguistic cues, a more rapid tone of voice resulted in a shorter time to recognize the emergency while a slow and calm voice increased the recognition time [10].

In this sense, our study contrasted the urgent voice with the calm voice by adjusting the paralinguistic cues focusing on the pitch and speech rate.

While an urgent voice tone is essential for effective evacuation, limited research has examined its impact on human-agent interaction, focusing on how users perceive its appropriateness. Previous studies have explored the context-appropriate tone of voice agents in daily task assistance [19], learning feedback [11], and notification systems [13], while its application in an emergency remains unexplored. This study investigates the emotional expression of a voice agent in a fire emergency setting, where users heavily rely on the agent's voice instructions to evacuate the situation.

Our study also focuses on the impact of the agent's voice tone on users' perceived trust as a primary indicator for evaluating smart agents [7, 18]. Trust, as a multilayered concept, requires a reliable and authoritative scale to measure. Trust can be categorized into different types according to the theory of reasoned action (TRA) [6]: "cognitive trust" is based on rational choices of the trustor motivated by the calculation of advantages, while "emotional trust" relies on the perceived affection, security, and comfort. Since the two kinds of trust can both affect the decision-making of the trustor but with different reasoning, it is necessary to include both types of trust when measuring the level of trust between a voice agent and the user.

Based on the investigations, we hypothesize that the agent's use of an urgent voice will promote users' effective evacuation from the emergency, compared to the agent using a calm voice for the instructions (H1). We also expect that users who received instructions in an urgent voice from the agent will show a higher level of cognitive and emotional trust in the agent (H2) and evaluate the agent's instruction as context-appropriate (H3) compared to those who interacted with the agent using a calm voice.

H1. *The agent's urgent tone of voice will promote users' effective evacuation from an emergency compared to a calm voice.*

H2. *The agent's urgent tone of voice will increase users' perception of (a) cognitive trust and (b) emotional trust in the agent compared to a calm voice.*

H3. *The agent's urgent tone of voice will increase users' perceptions of context-appropriateness compared to a calm voice.*

2 METHOD

2.1 Overview

To examine how the agent's voice tone affects the users' perceptions and behaviors in an emergency, we implemented an online simulation game where participants can experience a fire emergency interacting with a voice agent. A simulation game is a widely used experimental device that enables participants to be immersed in the evacuation scenario without exposing themselves to the real danger [14, 22]. We asked participants to perform the game in a desktop setting and complete a series of tasks exploring a confined virtual space. In order to evacuate the fire successfully, participants were guided and instructed by a voice agent with either a

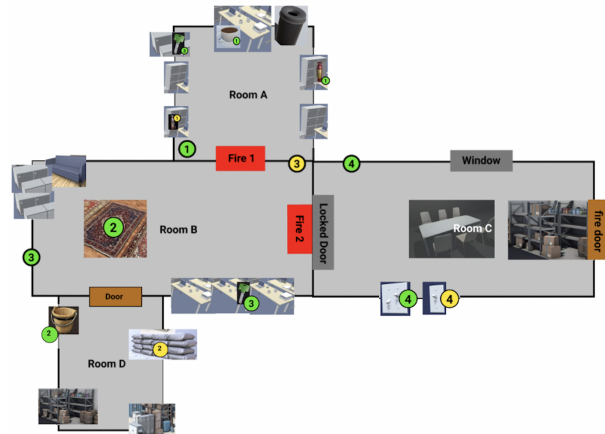


Figure 1: Early-level design along with the associated assets. During the simulation, participants begin in Room A where they face a Fire 1 blocking the doorway to Room B. Once they successfully extinguish Fire 1, they must proceed to Room D to get a sandbag, which will be used to put out Fire 2. After extinguishing Fire 2, participants can proceed to Room C, where they will break a window to safely exit the building.

"calm" or "urgent" tone of voice instruction. Two voice styles were pre-recorded by humans and distinguished by their differences in speed, pitch, intonation, and emphasis: the "calm" voice used a slow and steady tone with static intonations; whereas the "urgent" voice used fast and high-pitched tone with variable intonations at an inconsistent speed. During the game, participants' behavioral data were tracked to measure evacuation time. Then, participants were prompted to fill out a post-study survey to evaluate the voice agent's trustworthiness, and context-appropriateness.

2.2 Participants

Thirty participants were recruited through a university's recruiting systems and compensated with course credit for participating in our online experiment ($F=15$, $M=15$). Participants ranged from 18 to 38 years old with a mean age of 21 years ($SD=3.57$). Participants' degree of experience with voice agents such as Alexa or Siri ranged from individuals with no experience to daily users. Around 80% of the participants reported their native language as English. Participants were randomly assigned to one of the two experimental conditions: calm ($n=14$) and urgent ($n=16$).

2.3 Apparatus

The simulation game was built using Unity and WebGL for deployment. Assets for the game environment were found on the Unity Asset store. The image assets, which participants interacted with and stored in their itinerary, were found on Google Images. In addition, participant data was stored using Firebase's Firestore database. The game UI was initially brainstormed and created on Figma, and deployed on GitHub Pages.

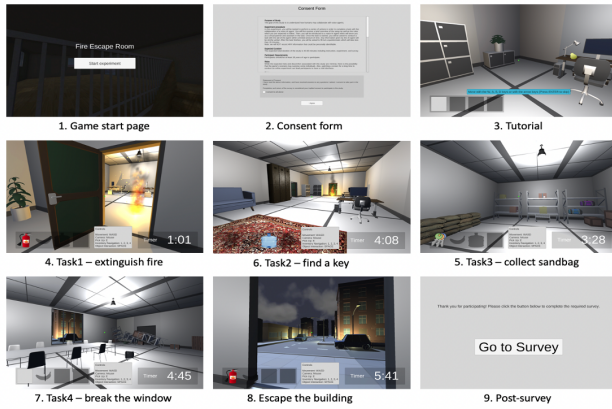


Figure 2: Experimental task procedure

2.4 Materials

2.4.1 Game design. In the game, a virtual office was designed allowing participants to navigate the space and interact with the items. The goal of this game is to evacuate from the virtual room by using the items instructed by the voice agent. Participants can use arrow keys for movement and change their viewing angle ranging 360 degrees by dragging with the mouse. Also, they can collect items by pressing the key "E" and use them in the location by pressing a space bar. All these control functions were instructed before starting the main tasks (see Procedure). To ensure every participant is aware of the game controls, the basic instructions for movement and interactions were displayed on the screen during the whole interaction.

The game provides a total of four tasks including (1) finding a fire extinguisher in one of the four cabinets and using it to put out a fire; (2) extinguishing another fire in front of the door with a sandbag; (3) opening the locked door with appropriate key; (4) smashing the window open with heavy objects, such as previously used extinguisher. Each task a participant had to complete had two or three false answers. A map of the game (Fig. 1) displays four rooms along with the locations of false options (in green) and true answers (in yellow) to evacuate the space appropriately. Participants can voluntarily determine to choose the options, while the voice agent would provide guidance for the right options.

2.5 Procedure

Participants were provided with a link that directed them to a consent form and the game under the agreement (Fig. 2). When the game starts, the voice agent introduced itself and began a tutorial to familiarize participants with the game control. Participants were required to collect three fruits around the virtual office by using the keyboard and mouse. Once they complete the tutorial, a fire emergency scenario was suddenly prompted, asking participants to escape the virtual office environment, which makes them more immersed in the unexpected emergency scenario. The urgency of evacuation was enhanced by the use of an in-game stopwatch.

In order to put out the fire and evacuate from the building, the voice agent provided instructions that were either delivered in an urgent voice or a calm voice. After completing the tasks, participants

Table 1: MANCOVA Results: Effects of Voice Agent's Emotional Tone on Dependent Variables

	Calm voice (n=14)		Urgent voice (n=16)		F(1)	p
	M	SD	M	SD		
Cognitive Trust	5.16	1.17	5.31	1.1	4.03	.04*
Emotional Trust	5.14	1.41	5.13	1.78	1.13	.29
Context-Appropriateness	3.86	1.55	4.44	1.76	5.67	.02*

Note. * $p < .05$. (2-tailed)

were asked to fill out a post-survey to share their perception and experience of the voice agent's instructions. The survey included questions related to the participant's overall experience with the agent, as well as their perception of the agent's tone of voice.

2.6 Measures

2.6.1 Behavioral data. In emergency situations, the goal is to evacuate as quickly as possible to ensure safety. To measure the participants' evacuation time, we collected the game log data to track when they started the game and completed each task. The game log included the completion time of the tutorial, the timestamp of the user's task actions, the completion time of each sub-task, and the completion time for the whole task measured in seconds. The evacuation time was calculated as the time elapsed from the completion of the tutorial to the completion of the whole task.

2.6.2 Self-report data. In the post-task survey, most questions used a seven-point Likert scale, with radio buttons for the participants to indicate their responses, anchored by "Strongly disagree" (=1) and "Strongly agree" (=7).

Both measures for cognitive and emotional trust were adapted from the previous literature about the trust of recommendation agents [12]. Cognitive trust ($\alpha=.79$) was evaluated with five items including the agent's expertise, knowledge, unbiased, honesty, and integrity. Emotional Trust ($\alpha=.96$) was evaluated based on three items regarding participants' sense of security, comfort, and content.

Context-appropriateness ($\alpha=.93$) was evaluated by asking about the perceived sense of emergency and appropriateness (i.e., "The tone of the agent's instruction conveys a sense of emergency;" "I felt the tone of the agent's instruction was appropriate in the scenario.") adapted from the questionnaire of perceived empathy [16].

Along with the questions about the dependent variables, we asked questions of control variables; to what extent participants took the instruction seriously, were motivated to evacuate, and understood the instruction [15].

3 RESULTS

We first compared the participants' evacuation time between the conditions by conducting an Analysis of Variance (ANOVA). However, no significant difference was found in the evacuation time between the calm condition ($M=420.7$, $SD=280.3$) and urgent condition ($M=520.65$, $SD=373.5$), $F(1, 13)=.67$, $p=.41$, rejecting the H1.

Then, we exploited Multivariate Analysis of Covariance (MANCOVA) to analyze cognitive trust, emotional trust, and context-appropriateness, considering covariate variables (Table 1). Confirming H2-a, we found that participants of urgent voice condition ($M=5.31$, $SD=1.1$) reported significantly higher cognitive trust

about the agent than those who use the calm voice agent ($M=5.16$, $SD=1.17$), [$F(1)=4.03$, $p=.04$]. However, there was no significant difference in emotional trust between the calm condition ($M=5.14$, $SD=1.41$) and the urgent condition ($M=5.13$, $SD=1.78$), [$F(1)=1.13$, $p=.29$], rejecting H2-b. There was a significant difference in context-appropriateness between the calm condition ($M=3.86$, $SD=1.55$) and the urgent condition ($M=4.44$, $SD=1.76$), [$F(1)=5.67$, $p=.02$]. Therefore, H3 was accepted showing that participants tend to feel the urgent voice is more appropriate to the context of emergencies than the calm voice.

4 DISCUSSION

Our study findings demonstrate that participants evaluated the voice agents using an urgent voice as trustworthy compared to the agent with a calm voice in the context of an emergency simulation. Also, individuals perceived urgent voice as context-appropriate to be used in an emergency scenario. These findings, building upon the previous body of research, indicate the importance of paralinguistic components of speech that vary users' experience and evaluation of human-agent interaction. In particular, the results of our study suggest that the manner in which a voice agent presents information to a user should depend on the context where that interaction occurs. For example, in the case of non-emergency interactions [13, 19], context-different emotions such as urgency and sternness have been demonstrated to negatively impact an individual's perception of an agent.

Our result highlights that the agent's emotional expression of urgency can particularly build cognitive trust in the agent's instruction. While trust is one of the most desirable components for voice agents which make guidance for humans' decision-making, its complexity of construct made difficulties in a clear investigation of trust in human-agent interaction. Acknowledging that, we measured trust divided into emotional and cognitive aspects and found that the agent's voice emotion can affect people's cognitive trust while having little impact on emotional trust. This result implies that the agent's expression of urgency in an emergency may lead to an increased perception of social intelligence and situational awareness, which are the key components for trust building [17]. It also aligns with the previous research that situation awareness is especially important in a high-risk context, such as the assistive robot for soldiers in jeopardy [4]. Since the agent's urgent voice did not change emotional trust in the agent, displaying a context-matching emotion may impress users with a sense of intelligence rather than affective and socially connected perceptions.

Trust can be measured by users' behaviors reflecting their psychological trust level during the interaction [1, 20]. While the agent's expression of urgency did not affect the users' evacuation time, participants in both calm and urgent condition groups had successfully evacuated from the virtual building assisted by the agents' instructions. That is to say, regardless of the agent's voice tone, the users followed what the agent instruct them to do and tried to evacuate the emergency as effectively as possible. On the topic of real-world applications, myriads of organizations ranging from department stores to hospitals and large-scale manufacturing facilities leverage voice agents to improve the human decision-making process. If such agents are to become a fixture in our society, it is

imperative to look into how people perceive the agents and how to build a better relationship with the agents according to the given tasks and contexts.

Participants in our study represent a selected group of people within the university student population. Nevertheless, our study findings are applicable to other demographic groups, given the nature of humans' universal desire to evacuate from risks and emergencies. One possible barrier of generalization is the users' ability to play the simulation game to be immersed in the emergency. Although our study statistically controlled how seriously participants consider the game and the agent's instructions to evacuate from the emergency, different level of skillfulness to play the game may vary their experience of participation.

Besides, there are a few aspects of voice agent design that could be further improved. The voice agent we designed only has two variants of emotional tones, either calm or urgent, and voice lines were recorded in a female voice. Potential improvements include increasing the type of emotions and integrating other types of gender voices to explore what different impacts they could have on the participants. Based on the discussion of nonverbal cues that manipulate different types of emotions, future work could also investigate the influence of both verbal and non-verbal expressions of an agent in an emergency, as well as a broader set of contexts for interaction beyond the emergency scenario.

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